



FUEL FORGE / PULSECHAIN

Protocol White Paper

Put your \$FUEL to work.

PLS reward cycles. wBTC Foundry rewards. FUEL, MORE and PAMP buy-and-burn.

VERSION 1.0 - 30 SEPTEMBER 2026 / PRE-LAUNCH DRAFT

Describes the current PulseChain contract candidate and agreed starting settings. Launch is delayed; no confirmed Fuel Forge deployment addresses or replacement launch time are published in this paper. The public design preview uses sample data and cannot transact.

thefuelforge.com/pulsechain-preview/

01 What Fuel Forge does

Fuel Forge uses FUEL in two distinct ways. A Forge position locks principal and may permanently burn extra FUEL for more power. A Foundry NFT permanently burns 100 million FUEL and receives an equal share of its mint cycle's wBTC participant pool. These are separate products; Foundry principal is never returnable.

PulseChain has chain ID 369. PLS is the native currency for fees, gas, Forge rewards and caller rewards. WPLS is used inside swaps. The Foundry reward asset is Ethereum-bridged wBTC, not Robinhood Chain cbBTC. Assets, positions and reward balances do not migrate automatically between chains.

Component	Starting design
Forge position	Positive FUEL principal; 8-1,000 day term; optional burn up to 3x principal
PLS reward vaults	8-, 28- and 88-day global cycles; power-weighted allocation
Foundry NFT	100 million FUEL burned per NFT; 288-day mint-cycle wBTC rewards
Batch mint	1-10 Foundry NFTs per transaction; fee and burn multiplied by quantity
Buy-and-burn	FUEL, MORE and PAMP; ten-minute intervals; user-triggered execution

Architecture and release scope

The candidate consists of ForgePosition, three PulsePositionRewardVault contracts, PulseFoundry, PulseFeeRouter, a one-time configured fee forwarder, three ForkBurnEngine contracts, swap adapters and quote contracts, and a SettlementBatcher. The external tokens already exist; Fuel Forge does not issue FUEL, MORE, PAMP or wBTC.

The website submits transactions to contracts after deployment verification. The design preview only demonstrates interactions. Mints, withdrawals, burns, settlements and claims need submitted transactions; no automatic keeper is required or guaranteed.

STATUS

This is an unaudited pre-launch specification. Confirmed deployment receipts, addresses, launch timestamps, current settings and route liquidity must be checked before wallet actions are enabled.

02 Forge principal, burn and power

Create a position by locking a positive amount of FUEL for a whole-number term of 8 through 1,000 days. Optional extra burn can range from zero to three times principal. The locked and burned amounts move in the same transaction. Additional burn is permanent; locked principal is potentially returnable after maturity.

INITIAL POWER

Power = principal + floor(principal x (termDays - 8) / 992) + burned FUEL. The duration bonus ranges from zero to one principal; the burn bonus adds FUEL one-for-one. Initial power cannot exceed five times principal.

100 million FUEL principal	Initial power
8 days / no extra burn	100 million (1x)
1,000 days / no extra burn	200 million (2x)
1,000 days / 100 million extra burn	300 million (3x)
1,000 days / 300 million extra burn	500 million (5x)

Power is an accounting weight attached to the position, not a separate transferable token. More power increases the position's share relative to other eligible power; it does not guarantee a larger currency payout if fee inflows or competing positions change. Extra burn and term do not increase the principal-based mint fee.

Maturity, grace and decay

Maturity is creation time plus the selected number of elapsed 24-hour days. There is no early principal withdrawal. For seven days after maturity, principal and power remain intact. Over the following seven days they decline linearly to zero, with principal and power rounded down separately.

Withdrawing during decay returns only the remaining principal and burns the rest. At fourteen days late, all principal is lost; anyone may finalize an expired position. Closing burns the position NFT but records the final owner for previously earned claims. Later decay does not reduce rewards already allocated at earlier deadlines.

Transferring a live Forge NFT changes control of remaining principal and unclaimed reward rights. It does not reset the term, maturity, burn or power.

03 PLS payout cycles

Three globally scheduled pools collect PLS and allocate it at 8-, 28- and 88-day closes. A position qualifies if created strictly before the deadline, still open at that deadline, and carrying positive historical power then. Any allowed lock term may qualify for any cadence under these rules.

Destination	Share of each Forge fee	Cadence
8-day PLS vault	13.12%	8 days
28-day PLS vault	11.07%	28 days
88-day PLS vault	8.61%	88 days
Foundry wBTC pool	8.20%	288 days

These reward allocations total 41% of a Forge fee. Anyone can settle a closed PLS cycle in bounded batches. Settlement reserves 0.25% of gross PLS for callers; the participant pool is the other 99.75%. Each claim is $\text{floor}(\text{participant pool} \times \text{position power at deadline} / \text{total eligible power})$.

An empty eligible set rolls participant funds into the then-current cycle. Claims do not expire in the current candidate. Anyone may trigger a claim, but payment goes to the position's current NFT owner or its recorded final owner after closure. Rounding dust rolls forward after all eligible claims.

Closes at 1 PM Eastern

Pulse-specific scheduling keeps reward closes at 1 PM America/New_York local time, including daylight-saving changes. The original plan was a 2 PM Eastern launch with first rounds ending at 1 PM on the calendar dates 8, 28, 88 and 288 days later. Those first rounds are shorter in wall-clock time; UTC elapsed length also reflects DST changes.

Launch has been delayed. A new launch date and anchor must be configured and verified before deployment; the old September 30 countdown is not proof of launch. Burn intervals start from the actual launch timestamp and remain ten elapsed minutes. No final close dates are promised here.

HYPOTHETICAL PAYOUT

If a PLS cycle contains 10 million PLS gross, callers reserve 25,000 PLS and participants share 9,975,000 PLS. A position holding 5% of eligible power receives 498,750 PLS before rounding. Actual funding and competing power are unknown in advance.

04 Foundry NFTs and wBTC

Each Foundry NFT burns 100 million FUEL and pays a separate PLS mint fee. A ten-NFT batch burns one billion FUEL and pays ten per-NFT fees in one transaction. Quantity multiplies the clamped per-NFT fee; it does not apply the floor only once to the whole batch. Gas is additional.

NFTs minted during the same 288-day cycle share that cycle's wBTC participant pool equally. A Foundry NFT participates in its mint cycle only, not automatically in every later cycle. It remains transferable, carrying any unclaimed reward entitlement from that cycle.

The starting Foundry fee is 10% of the quoted PLS value of 100 million FUEL, clamped to 2 million-20 million PLS per NFT. Of the paid Foundry mint fee, 1% goes to development and 99% purchases wBTC. Separately, the Foundry receives the router's 8.20% share of Forge fees for wBTC purchases; those routed receipts are converted in full.

At a cycle close, anyone may settle. The caller receives 0.25% of gross wBTC and eligible NFTs split the remaining 99.75%. With no eligible NFTs, participant funds roll forward; residual token-unit dust rolls forward after all claims. Current NFT owners receive unclaimed rewards.

Conversion limits

The starting conversion cap is 20 million PLS per swap. A large batch may need multiple swaps. Failed swaps may retry smaller protected chunks within the same transaction, up to 32 completed swaps and 64 attempts. Quote failures, insufficient output or exhausted limits revert the whole mint or conversion; a reverted mint does not burn FUEL or issue NFTs.

The owner can change the conversion cap and lower slippage from the 10% ceiling. Splitting trades does not eliminate price impact. The selected direct WPLS/wBTC route uses PulseX V2; routing and reserves must be rechecked at launch.

HYPOTHETICAL FOUNDRY FEE

If 100 million FUEL is quoted at 25 million PLS, 10% gives 2.5 million PLS per NFT. A two-NFT mint burns 200 million FUEL and pays 5 million PLS; 50,000 PLS goes to development and 4.95 million PLS purchases wBTC, subject to execution.

05 Mint fees and routing

Forge starts at 100% of the quoted PLS value of locked principal, bounded to 200,000-200 million PLS per position. Foundry starts at 10% of the quoted value of its fixed burn, bounded to 2-20 million PLS per NFT. These are independent owner-controlled policies, not immutable fee promises.

FEE CALCULATION

Calculate the percentage of the oracle's quoted FUEL value, then clamp it between the configured minimum and maximum when bounds are enabled. The owner can set each percentage above zero through 100%, change the floor and cap, or disable bounds independently for Forge and Foundry.

Forge fee destination	Starting share
8-day PLS rewards	13.12%
28-day PLS rewards	11.07%
88-day PLS rewards	8.61%
Foundry wBTC purchases	8.20%
FUEL buy-and-burn	28.00%
MORE buy-and-burn	20.00%
PAMP buy-and-burn	10.00%
Development	1.00% nominal, plus rounding dust

Rewards stay fixed at 41% and burns at 58% total. The owner may change FUEL, MORE and PAMP allocations together, provided they total 58% of all Forge fees. These percentages describe gross protocol-fee shares. An engine's daily drip is separate and does not automatically change when its fee share changes.

Forge fee routing is atomic: a required receiver failure reverts position creation. Foundry mint fees take their own 1%/99% path and are not redistributed through the Forge fee router. The public site should display confirmed on-chain shares and fee policies rather than treating preview values as live.

06 Buy-and-burn operations

FUEL, MORE and PAMP engines receive PLS through the fee router. Each starts at a 1% daily drip spread across 144 ten-minute intervals. The owner independently controls each drip from 1% to 10%, its positive gross execution cap, and slippage within the 10% ceiling.

The starting gross cap is 100,000 PLS per execution for each engine. Unexecuted interval weight accumulates. A cap can leave executable value for later calls; rate changes checkpoint weight accrued under the previous rate. The current balance, elapsed weight, existing carry and cap determine executable value, so this is not a fixed monthly token purchase commitment.

A successful caller receives 1.5% of gross PLS and the other 98.5% buys the relevant token, sent to the dead address. The caller pays gas; small rewards may not cover it. A failed swap pays no caller reward and burns no tokens, although a reverted on-chain transaction still costs gas.

ILLUSTRATIVE TEN-MINUTE INTERVAL

With 1 million PLS in a pool, a 1% daily drip and one newly elapsed interval, gross executable value is about 69.4444 PLS. The caller receives about 1.0417 PLS and about 68.4028 PLS buys tokens. With 144 intervals accrued and an unchanged starting balance, gross is 10,000 PLS before caps and prior executions.

Price references and safeguards

FUEL fee values and swap references come from current executable PulseX V2 router quotes for the selected pools. The fee oracle quotes principal using a pool-based unit FUEL price; it is a valuation for fee calculation, not proceeds guaranteed from selling a large amount. USD figures on the website are informational and do not set on-chain fees.

Burn and Foundry quotes must be valid and no more than fifteen minutes old. Minimum output guards apply the configured shortfall limit. A fresh quote from the same pool is not an independent fair-price oracle and may already reflect a manipulated or thin market. Liquidity changes can cause poor output or reverts even when earlier fork tests passed.

The starting routes are direct WPLS/token pairs on PulseX V2. Do not use the earlier superseded FUEL token/pool or the DAI/wBTC route as the configured direct Foundry route.

07 Owner powers and risks

The intended owner and development wallet is 0x02A0d741FBaebC03A8f0d1A85670bf1CA8C15fA9. The owner can change Forge and Foundry fee policies independently, the distribution of the fixed 58% burn budget, each burn engine's drip/cap/slippage, and Foundry swap cap/slippage. The fee-policy code has no timelock.

Reward-router allocations remain fixed. Swap adapters are configured once. The inspected candidate does not expose an owner withdrawal of locked Forge principal or allocated participant rewards. These observations are not an independent security assurance.

Risk	Practical consequence
Contract or integration error	Lost or trapped assets, failed transactions or incorrect accounting; no independent audit.
Thin or manipulated liquidity	Distorted entry fees, material price impact, low permitted outputs or swap failures.
Token and bridge risk	FUEL, MORE, PAMP, PLS and wBTC can lose value. Bridged wBTC adds bridge, custody and underlying-token dependencies.
Irreversible burn / late decay	Optional burn and Foundry burn cannot be recovered. Late Forge principal can decay to zero.
Owner-controlled settings	Entry fees and operating limits may materially change after this PDF is published.
Manual execution	Settlement, claims, burns and withdrawals depend on submitted transactions and gas.
No fixed yield	Payouts depend on actual inflows and eligible shares. Burns do not guarantee appreciation.

Testing status

The September 30 local Solidity run passed 134 tests. Enabling previously gated fork groups produced 39 passing PulseChain checks and 46 passing Robinhood checks, with no remaining failures after correcting an outdated Robinhood fee expectation. PulseChain checks included current routes, burn backlogs, a 20-million-PLS wBTC conversion and ten-NFT stress conversion.

These were local simulations with seeded wallet/token balances where needed; they did not broadcast transactions or establish that users can obtain the tested FUEL quantities. Quotes and liquidity differ across fork snapshots and can change before launch. A clean test run does not constitute an audit or a deployment approval.

08 Token and route reference

These are configured external assets and pools, not confirmed Fuel Forge deployment addresses. Recheck bytecode, token order, decimals, factory, router, reserves and executable output on chain 369 before launch.

Token	PulseChain address
WRAPPEDNATIVE	0xA1077a294dDE1B09bB078844df40758a5D0f9a27
FUEL	0x6633aeDbB64115391238D7ce56D2DFAa2e0c7b45
MORE	0xbEEf3bB9dA340EbdF0f5bae2E85368140d7D85D0
PAMP	0xd5E952eA1B17034Ad3368805ed9CFB08009deA79
wBTC	0xb17D901469B9208B17d916112988A3FeD19b5cA1

Direct WPLS pair	Pool address
FUEL	0x0bb20331f424e59612668f3294a23caa83cc06ef
MORE	0x3D3B080A1Ec1AFc121a27AE4cBad17A14E80f7B5
PAMP	0x5A6ed52a40983BDE815Cffe445F2175E80456F6B
wBTC	0x8c52470a05eEB2fCe4905688Ec59bFDd32E71D07

Infrastructure	Address
PulseX V2 router	0x165C3410fC91EF562C50559f7d2289fEbed552d9
PulseX V2 factory	0x29eA7545DEf87022BAdc76323F373EA1e707C523

PROTOCOL REGISTRY PENDING

Forge, Foundry, reward vault, fee-router, burn-engine, adapter and settlement-helper deployment addresses must be added after successful receipts and public verification. No local-fork or dry-run addresses are official deployments.

Primary references and versioning

Prepared from ForgePosition.sol, ForgeMath.sol, FeePolicy.sol, PulsePositionRewardVault.sol, PulseEasternSchedule.sol, PulseFoundry.sol, ForkFoundry.sol, PulseFeeRouter.sol, ForkBurnEngine.sol, PulseV2RouterQuote.sol, PoolFuelPriceOracle.sol, ForkSwapAdapter.sol and DeployPulseChainCandidate.s.sol, with the reviewed local PulseChain config and fork results.

Public interface: <https://thefuelforge.com/pulsechain-preview/> . Explorer: <https://scan.pulsechain.com/> . PulseChain RPC: <https://rpc.pulsechain.com/> . This dated draft replaces Robinhood references only on PulseChain pages. Update it after deployment with confirmed addresses, launch anchor, settings and a change log.